

THILAC RAMESH

Mechanical Engineering Undergraduate | Robotics & Autonomous Systems

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Profile

Mechanical Engineering undergraduate at the **University of Peradeniya** with interdisciplinary experience in **autonomous robotics, SLAM, ROS 2, mechanical design, finite element analysis, IoT systems, and engineering software development**. Currently contributing to autonomous navigation and robotic perception research using the **Unitree Go2 EDU** quadruped platform.

Strong interest in **robotics, autonomous systems, intelligent mechatronics, mechanical design, simulation, computer vision, and perception**, with experience integrating mechanical engineering, software development, AI, and rapid prototyping.

Education

B.Sc. Engineering (Hons) in Mechanical Engineering

University of Peradeniya

2024 – Present

Peradeniya, Sri Lanka

- Focus Areas: **Robotics & Automation, Machine Design, Mechanical Systems, and Intelligent Engineering Systems.**

Certifications

Certified SOLIDWORKS Associate (CSWA)

Dassault Systèmes

- Certified in fundamental **SOLIDWORKS 3D CAD** modelling, part design, assemblies, engineering drawings, and engineering design evaluation.

Experience

Research Team Member – Robotics

WSO2 – Research Apprenticeship

Jun 2026 – Present

Remote / Hybrid

- Developing and evaluating **SLAM optimization** and **BIM-integrated navigation** approaches for autonomous operation of **Unitree Go2 EDU** quadruped robotic platforms.
- Integrating **perception, localization, mapping, navigation, and real-time robotic data pipelines** for autonomous indoor robotic operation.
- Prototyping autonomous **patrol and navigation capabilities** within policy-based **ACSAR-E** robotic frameworks.

Technical Consultant – IoT Systems

MyDynamica – Freelance

Feb 2026 – Apr 2026

Remote

- Architected the **JPL IoT Middleware** using **FastAPI**, integrating physical IoT infrastructure with digital management systems.
- Designed a resilient database architecture incorporating **Change Data Capture (CDC)** and read replicas for real-time synchronization and security automation.
- Developed backend data pipelines for low-latency communication between IoT devices, databases, and management interfaces.
- Prepared technical documentation and provided **Design for Manufacturability (DFM)** guidance for hardware integration.

ICT Technician Trainee

Computer Resources Center

Oct 2023 – Mar 2024

Point Pedro, Sri Lanka

- Diagnosed and resolved hardware, networking, and system configuration issues across organizational ICT infrastructure.
- Supported network configuration, system maintenance, hardware inventory, and end-user technical troubleshooting.

Research

Higher-Dimensional Radial Model (HDRM) (*under review*)

Apr 2026

Independent Researcher

- Developed a classical geometric framework for investigating geometric relationships underlying quantum-mechanical probability distributions.
- Formulated a higher-dimensional radial representation to investigate geometric structures associated with probabilistic behaviour.

- Prepared and submitted a formal research manuscript to the *International Journal of Theoretical Physics* (Springer Nature).

Selected Engineering Projects

Two-Way Manual Meat Mincer – Planetary Gear Transmission

June 2026 – July 2026

Mechanical Design Project – University of Peradeniya

- Designed a manually operated **two-way meat mincer** incorporating a **planetary gearbox** and mechanical power-transmission system.
- Designed the sun, planet, and ring gear arrangement to obtain the required torque and relative rotational motion.
- Performed component sizing and structural calculations using **von Mises failure theory, shaft torsion, bending analysis, Lewis gear-tooth bending analysis, and engineering safety factors**.
- Designed the gears, shafts, auger, cutting mechanism, housing, handle, and suction-base mounting system using **SOLIDWORKS**.
- Considered manufacturability, ergonomics, material selection, food safety, structural reliability, and maintenance requirements.

Universal Mechanical Brace – Adaptive Wedge Orthosis

Feb 2026 – March 2026

Mechanical Designer

- Designing an adaptive orthotic mechanism using **tribological principles** to improve mechanical interaction, wear resistance, and adjustability.
- Applying **Finite Element Analysis (FEA), topology optimization, and motion analysis** in SOLIDWORKS to evaluate structural performance and reduce component mass.
- Investigating adjustable mechanical architectures for adaptable support while maintaining structural reliability and manufacturability.

Open-Source Software

ezclean – Python Library

2026

Publisher & Maintainer

- Published and maintain **ezclean**, an independently developed Python software library.
- Responsible for **package development, maintenance, documentation, version management, and public distribution**.

Technical Skills

Robotics & Autonomous Systems: ROS 2, SLAM, Autonomous Navigation, Unitree Go2 EDU SDK, MuJoCo, Robot Simulation, Perception Systems

Mechanical Design & CAE: SOLIDWORKS (CSWA Certified), Machine Design, Finite Element Analysis, Topology Optimization, Gear Design, Shaft Design, Failure Analysis, Engineering Materials

Programming & Software: Python, C, Python Package Development, FastAPI, Flask, Qt, Linux, Git, SQL, MariaDB

Artificial Intelligence & Computer Vision: Machine Learning, Computer Vision, NLP, Scikit-learn, spaCy, Sentence-BERT, Pandas, Matplotlib

Manufacturing & Prototyping: FDM 3D Printing, Rapid Prototyping, Slicer Optimization, Design for Manufacturability, Hardware Troubleshooting

3D Design & Visualization: Blender, 3D Modelling, 3D Rendering, Technical Visualization

Graphic Design & Adobe Creative Suite: Adobe Photoshop, Adobe Illustrator, Adobe Lightroom, Graphic Design, Image Editing, Vector Graphics, Visual Content Design

Engineering Documentation: LaTeX, Technical Reports, Research Papers, Project Proposals, Engineering Documentation

Competitions & Awards

NexusHacks 2026 – 2nd Place, Agentic AI Category

2026

NexusHacks – India

India

- Secured **2nd Place in the Agentic AI category** at NexusHacks 2026.

DataStorm 7.0 – Finalist

2026

Sponsored by Keells Octave

Sri Lanka

- Reached the **final stage of DataStorm 7.0**, a competitive data analytics and problem-solving competition.

- Received the **Special Award for Technical Design** for an AI-driven OSINT and socio-economic intelligence platform.
- Developed an automated analytics pipeline using **spaCy Named Entity Recognition, Sentence-BERT, K-Means clustering, VADER sentiment analysis, and TF-IDF** for semantic topic discovery, event analysis, and risk assessment.
- Built a **Flask-based backend architecture** integrating automated data processing, task scheduling, statistical event detection, and dynamic keyword extraction.